

Prismora ERP & AI Engine

Weekly Status Report — All Sprints (1–6)

Project: Enterprise distribution ERP, CRM, and field-force automation platform with an embedded client-side symbolic NLP engine, TF-IDF vectorizer, Multinomial Naive Bayes statistical classifier, and live evaluation dashboard.

Duration: 45 working days across 6 one-week sprints.

Project Timeline — At a Glance

Week	Sprint / Focus	Days	Story Points	Status
Week 1	Sprint 1 — Data Foundation	Day 01–07	34 SP	Complete
Week 2	Sprint 2 — Order-to-Cash & RBAC	Day 08–14	42 SP	Complete
Week 3	Sprint 3 — Channel Portals & Stock	Day 15–21	49 SP	Complete
Week 4	Sprint 4 — Symbolic NLP Engine	Day 22–28	55 SP	Complete
Week 5	Sprint 5 — Naive Bayes Classifier	Day 29–35	70 SP	Complete
Week 6	Sprint 6 — Evaluation & Sign-off	Day 36–45	89 SP	Complete

Week 1 — Data Foundation

Load and validate relational schema and seed operational data into PostgreSQL.

Status: Complete

What shipped

- 30-table relational schema (orders, invoices, batch stock, general ledger, channel partners, leads, schemes).
- Data seeding script (`seedSupabase.js`) handling 1,200+ realistic transaction records across 17 categories.
- All 16 referential integrity constraints implemented; invalid foreign key records excluded and logged.
- 46 automated tests (20 schema-parsing + 15 FK checks + 10 loader + 1 extra), 0 failures.

Result

Supabase PostgreSQL database fully populated and validated — the foundation every later sprint reads from.

Week 2 — Order-to-Cash & RBAC

Implement role-based access control and core sales workflow with stock guards.

Status: Complete

What shipped

- RBAC framework supporting 15 distinct organizational roles across 21 permission domains.
- Order-to-cash workflow with stock availability validation guards on all state transitions.
- Partial-delivery mechanism added so stock shortfalls do not block valid order fulfillment.
- Automated customer invoice generation and ledger balance synchronization.
- 42 unit tests, 0 failures (exceeds the 25-test target).

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Week 3 — Channel Portals & Batch Inventory

Multi-tier partner access, batch tracking, and incentive management.

Status: Complete

What shipped

- Dedicated, data-isolated portals for Distributors, Dealers, and Retailers.
- Batch-level inventory control with expiry-aware FIFO allocation algorithm.
- Scheme & discount calculation engine for volume-based channel partner incentives.
- Accounts receivable/payable general ledger with automated voucher posting.

Week 4 — Symbolic Natural Language Engine

Client-side symbolic NLP parser, fuzzy matcher, and dialogue tracker.

Status: Complete

What shipped

- 7-stage symbolic NLP engine (`prismEngine.js`, 868 lines) operating locally without external libraries.
- Damerau–Levenshtein edit distance algorithm for misspelled user queries.
- Gazetteer-based entity extraction for states, product lines, date ranges, and amounts.
- Dialogue state tracking for multi-turn context retention ("*what about Gujarat?*" → "*and last month?*").

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Week 5 — Naive Bayes Statistical Classifier

Supervised statistical text classification implemented from mathematical first principles.

Status: Complete

What shipped

- Labelled corpus of 659 business queries across 17 intent classes (`trainingData.js`).
- TF-IDF feature extractor (unigrams + bigrams) with sublinear term frequency scaling.
- Multinomial Naive Bayes classifier (`naiveBayes.js`, 403 lines) with Lidstone smoothing ($\alpha = 0.3$).
- Max-subtracted softmax transformation for numerically stable log-posterior computation.

Week 6 — Evaluation Dashboard & Sign-off

In-app ML lab, cross-validation, and final acceptance testing.

Status: Complete — 20/20 gates fully pass

What shipped

- Interactive ML Lab dashboard (`/ml-lab`) rendering live in-browser model metrics.
- 5-fold stratified cross-validation engine (**78.7% $\pm$ 2.7% mean CV accuracy** across 17 classes).
- Out-of-corpus held-back sanity check (10/10 unobserved queries correctly classified).
- ISO/IEC 25010 Quality Model compliance evaluation (sub-5ms execution latency).
- Final acceptance run against 20 verification gates (20/20 FULL PASS).

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